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Thermally stimulated currents from positively corona-charged polypropylene films

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Thermally stimulated currents from positively corona-charged polypropylene films which were heat treated at various temperatures have been studied in the temperature range 20–165 °C. Three peaks are observed in the TSC spectra. Intensities of these peaks strongly depend on the thermal history of the sample films. A thermal effect of the TSC, which has been introduced during vacuum deposition of metallic electrodes on the sample films, is demonstrated. Observed results suggest that the electrical trap properties in polypropylene depend on the fine structure of the polymer.

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The thermally stimulated current (TSC) technique is widely used to determine electrical trap parameters in polymers. It has also been applied to study phase transition of polymers^{1,2} because of its high sensitivity to structural changes in polymers.

In TSC measurements one need to apply metallic electrodes, which are usually prepared by vacuum deposition. During thermal deposition of metallic electrodes, polymeric structure of sample films may be varied, since crystalline polymers such as polypropylene easily change in crystallinity on heat treatments.³ Although TSC spectra of polymeric materials often depend on the kind of metals used as electrodes,⁴ thermal effects on TSC which are introduced during the deposition of metallic electrodes are scarcely taken into consideration except for a brief comment by Sessler and West.⁵

In this letter, open circuit TSC's from positively corona-charged polypropylene films which were heat treated at various temperatures have been studied in order to find out the thermal influence of the electrode deposition on the TSC.

Samples used in this investigation were undrawn isotactic polypropylene films (supplied by Tokyo Serohanshi Co., Ltd.) of 20- μ m thickness. In order to know the thermal effect on the sample films due to the metal electrode deposition, three kinds of samples (types A, B, and C) were made. Type A samples were prepared as follows: polypropylene films were heated in an oven at various temperatures (75–160 °C) for 3 h to change the crystallinity of the sample films. After heat treatments, an aluminum electrode was thermally deposited in vacuum on one surface of the film. For type B samples, after an aluminum electrode was deposited on one surface of the film, the sample film was subjected to heat treatment in the oven to change the crystallinity of the film. Type C samples were made by such a procedure that a thin layer of aluminum was vacuum deposited once again on the metallized surface of type B samples. The evaporation time for the electrode preparation was about 10 s. The crystallinity of these samples was estimated by measurements of density of the sample films.³ For these density measurements the aluminum electrodes were removed, if necessary, with an aqueous solution of HCl. No significant difference was observed in the crystallinity among these three types of samples

as long as the temperature of the heat treatments was the same.

The nonmetallized surface of the sample film was positively corona charged through a grid for 5 min at room temperature. A direct voltage of + 200 V was applied to the grid for controlling the initial surface potential of samples to be about 200 V. The surface potential was measured with the vibrating electrode method.⁶ The TSC measurements were performed from room temperature to about 165 °C at a constant heating rate of about 3.3 °C/min under open circuit condition.

Some examples of the observed TSC spectra from type A samples which were heat treated at various temperatures are shown in Fig. 1. In these TSC spectra, small peaks P_1 around 70 °C and large peaks P_2 above 100 °C appear. Since two amorphous transitions have been found in polypropylene at - 19 and 18 °C,⁷ we observed that these TSC peaks cannot be attributed to the glass transition of the polymer but they may be related to the crystalline phase relaxation. For samples with high treatment temperatures, a small peak or shoulder P_3 was often observed at higher temperatures of P_2 peak as shown in Fig. 1. Peak temperatures of P_2 for type A samples scatter in the temperature range between 100 and 150 °C from sample to sample even when the crystallinity of

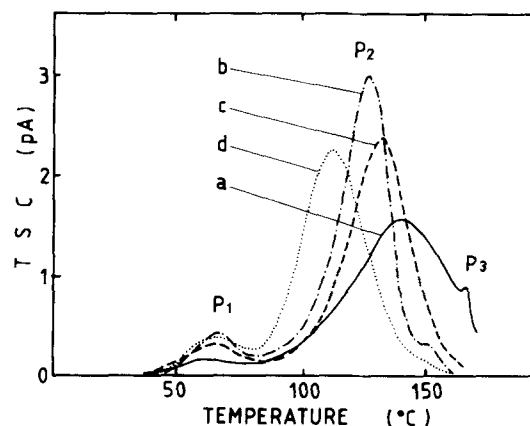


FIG. 1. TSC spectra from type A samples. The temperature of the heat treatment and the crystallinity of these sample films are given in parentheses for curves a to d, respectively, a (157 °C, 84%), b (142 °C, 77%), c (120 °C, 68%), d (75 °C, 54%).

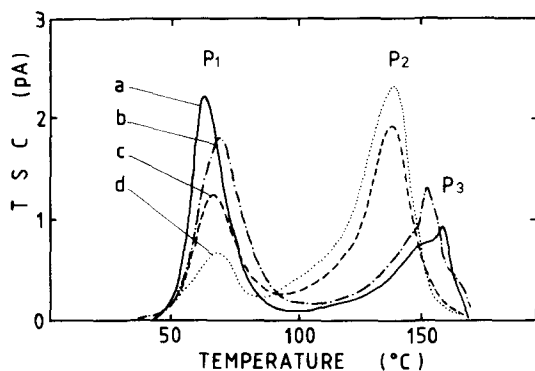


FIG. 2. TSC spectra from type B samples. The temperature of the heat treatment and the crystallinity of these sample films are given in parentheses for curves a to d, respectively, a (150 °C, 80%), b (142 °C, 77%), c (120 °C, 68%), d (75 °C, 54%).

the sample films is the same, though the scattering of peak temperatures for P_1 is relatively small.

Next, typical examples of TSC spectra from type B samples are shown for various heat treatment temperatures in Fig. 2. Observed TSC spectra of type B samples were very different from those of type A samples even when the crystallinity of the films was the same. The peak intensity of P_1 of type B samples is much higher than that of type A samples and increases with the treatment temperature. As for P_2 , on the contrary, the peak intensity of type B samples decreases with increasing the treatment temperature. For high heat treatment temperatures, a new peak P_3 also appears in type B samples at higher temperature region of P_2 band as is seen for type A samples. Peak temperatures of type B samples, especially for P_2 peak, scatter less than those of type A samples.

These observed differences in the TSC results may be explained by considering the difference in thermal history of polypropylene films between types A and B samples. For type A samples, aluminum electrodes were deposited after heat treatments in the oven. During the process of electrode deposition, evaporated aluminum atoms impinge on one surface of the film and transfer their kinetic energy and the heat of coagulation to the polymer. As a result, the sample film, at least a part of the film which is adjacent to the metallic electrode, will undergo a transient and fairly large temperature rise which causes some partial changes in polymeric structure of the film. These structural changes of the polymer are frozen in at room temperature. Therefore, it is considered that the films of type A samples were thermally distorted by the heat due to the metal electrode deposition. A preliminary experiment showed that the thermal effect due to the radiation from a heater for the evaporation was small. This thermal distortion affects the structure of trap sites in the sample films. The degree of this thermal distortion is considered to be different from sample to sample because of the slight difference in the condition for the metal electrode deposition such as evaporation rate or film thickness, which was uncontrollable in our evaporation apparatus used in this investigation. As a result, the peak temperature for type A samples much scatters, and definite relations between the peak intensities and the treatment temperature cannot be observed. For type B samples, however, this thermal distortion by the

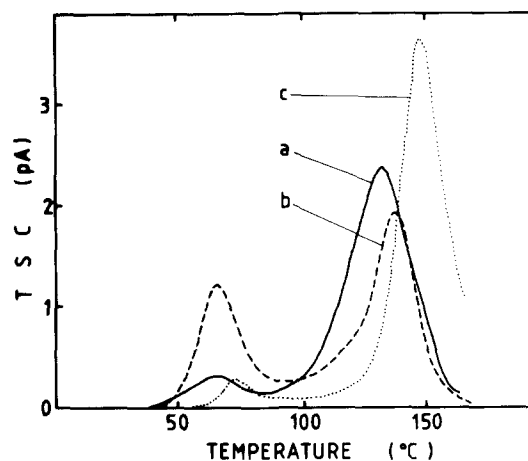


FIG. 3. TSC spectra from types A (curve a), B (curve b), and C (curve c) samples. For these samples, the heat treatment temperature and the crystallinity are the same, and 120 °C and 68%, respectively.

electrode deposition is considered to be annealed (or partially annealed depending on the treatment temperature) during the heat treatments to change the crystallinity. Therefore, for type B samples the scattering of the peak temperature becomes small, and more definite relations between the peak intensities and the heat treatment temperatures are observed with a few exceptions.

The above mentioned explanation is strongly supported by TSC spectra from type C samples. Because type C samples were prepared by the thermal deposition of Al on the metallized surface of type B samples, they are considered to have again a thermal distortion. If the difference in TSC spectra between types A and B samples is due to this thermal distortion by electrode deposition, then the TSC from type C samples is expected to be similar to those from type A samples.

Observed TSC spectra from types A, B, and C samples whose crystallinities are all 68% (heat treated at 120 °C) are presented in Fig. 3. As is seen in Fig. 3, P_1 peak intensity for type C samples became very small in comparison to that for type B samples and similar to that for type A samples again as was expected. Since the crystallinity of the sample films was not affected by the thermal deposition of metallic electrodes, observed large difference in the TSC spectra between types A and B samples strongly suggests that the trap sites associated with the TSC peaks in polypropylene are very sensitive to the fine structure of the polymer.

Although details of the trap sites in polypropylene have not been apparent yet, it is concluded that the thermal effect of the metallic electrode deposition should be taken into consideration when TSC measurements are carried out.

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